

PAPER_1112: Production Scaling V26 Pipeline — UQFF Whitepaper Generation Framework

Star Magic UQFF Framework

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Abstract

We document the V26 production pipeline for UQFF whitepaper generation, covering the 26-dimensional vacuum density scaling framework, batch PDF rendering via pandoc+XeLaTeX, arXiv-style LaTeX compliance, and CVW v2.0.0 gate enforcement. The pipeline ensures all 1,000 target whitepapers adhere to the canonical UQFF equations from `scm_{vacuum\_manifold}.py`, use consistent arXiv LaTeX formatting (no mixed formats), and have corresponding PDFs in `pdf/`. Progress tracking integrates with the PAPER_001–1141 suite.

1. V26 Pipeline Architecture

The production pipeline operates in six stages:

1. **Content generation:** Python scripts create `PAPER_{N_Title}.md` in `whitepapers/`
 2. **LaTeX compliance check:** All math must use `$...$` or `$$...$$` (no Unicode math, no `\(...\)`)
 3. **CVW v2.0.0 gate:** G1–G6 gate checks enforce canonical UQFF constants
 4. **PDF rendering:** `pandoc --pdf-engine=xelatex -V geometry:margin=1in -V fontsize=11pt -V colorlinks=true`
 5. **Gap audit:** Compare `whitepapers/PAPER_*.md` vs `pdf/PAPER_*.pdf`
 6. **Git commit:** `git add -A ; git commit -m "docs: ..." after each batch`
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2. Canonical UQFF Constants (V26 Mandatory)

All whitepapers must use these exact values:

$$\begin{aligned}\rho_{\text{vac,SCm}} &= 7.09 \times 10^{-37} \text{ J/m}^3 \\ \rho_{\text{vac,UA}} &= 7.09 \times 10^{-36} \text{ J/m}^3 \\ [SSq] &= 0.57, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1} \\ \beta_i &= 0.6, \quad F_{TRZ} = 0.1, \quad \Phi_{\text{res}} = 0.84 \\ S_{26}^{(3)} &= 1.4531 \times 10^{26}, \quad f_{\text{THz}} = 1.25 \text{ THz}\end{aligned}$$

3. VDS Master Equation

$$\text{VDS}([SSq]) = \sum_{n=1}^{\infty} \frac{[SSq]^n}{n^{26}} = \text{Li}_{26}(0.57)$$

The Ramanujan order-3 acceleration operator $S_{26}^{(3)}$ amplifies the raw Li_{26} value to the physical scale 1.4531×10^{26} .

4. F_{U_Bi_i} Master Equation

$$F_{U,Bi,i} = \int_0^\infty \left(-F_0 + \frac{GM}{r^2} + \rho_{\text{vac,SCm}} \cdot U_{UA} \cdot \cos(\pi t_n) \right) dr$$

with $t_n < 0$ (negative-time resonance gate) and $F_0 = 1.0 \times 10^{-10} \text{ N/m}^3$.

5. Holmlid KER Verification

Every pipeline run verifies the canonical KER computation:

$$E_{\text{KER}} = h \cdot f_{\text{THz}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} = 630 \text{ eV}$$

$$h \cdot f = 6.626 \times 10^{-34} \times 1.25 \times 10^{12} = 8.283 \times 10^{-22} \text{ J}$$

6. CVW v2.0.0 Gates

Gate	Description
G1	All ρ_{vac} values use 7.09×10^{-37}
G2	$[SSq] = 0.57$ exactly
G3	$S_{26}^{(3)} = 1.4531 \times 10^{26}$
G4	Negative-time: $t_n < 0$, uses $\cos(\pi t_n)$
G5	No Standard Model gravity (GM/r^2 is UQFF emergent only)
G6	All arXiv LaTeX: $\$ \dots \$$ or $\$ \$ \dots \$ \$$ only

References

1. Canonical SCm equations: `scm_{vacuum\_manifold}.py`
2. PDF generation: `generate_pdfs.ps1`
3. CVW compliance: `.github/copilot-instructions.md`

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic